LOGISTIC REGRESSION

For each observation in our dataset, we have

input: $\underline{x} \in \mathbb{R}^d$ $\underline{x} = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_d \end{bmatrix}$, $a_i \in \mathbb{R}$

output: $y \in \{1, 2, \dots, K\}$ → K distinct classes.

We want to model $P(y = k \mid X = x)$ as:

$p_k(\underline{x}; \theta_k)$ where θ_k are the parameters of our model. More specifically, we model:

$p_k(\underline{x}; \theta_k) = \sigma(\underline{x}\theta_k)$ where σ is the softmax function and $\theta_k \in \mathbb{R}^d$ $(\mathbb{R}^{d+1})$.

$$\Rightarrow p_k(\underline{x}; \theta_k) = \frac{e^{\underline{x}\theta_k}}{\sum_{i=1}^{K} e^{\underline{x}\theta_i}}$$

We can denote $\underline{\theta} = [\theta_1, \theta_2, \dots, \theta_K]$ and have:

$$p(\underline{x}; \underline{\theta}) = \frac{1}{\sum_{i=1}^{K} e^{\underline{x}\theta_i}} \begin{bmatrix} e^{\underline{x}\theta_1} \\ e^{\underline{x}\theta_2} \\ \vdots \\ e^{\underline{x}\theta_K} \end{bmatrix}$$

Notice that the elements of this vector are all positive and add up to 1!

So we can think of them as pseudo-probabilities.

How do we determine $\underline{\theta}$? We minimise the cross-entropy loss function.

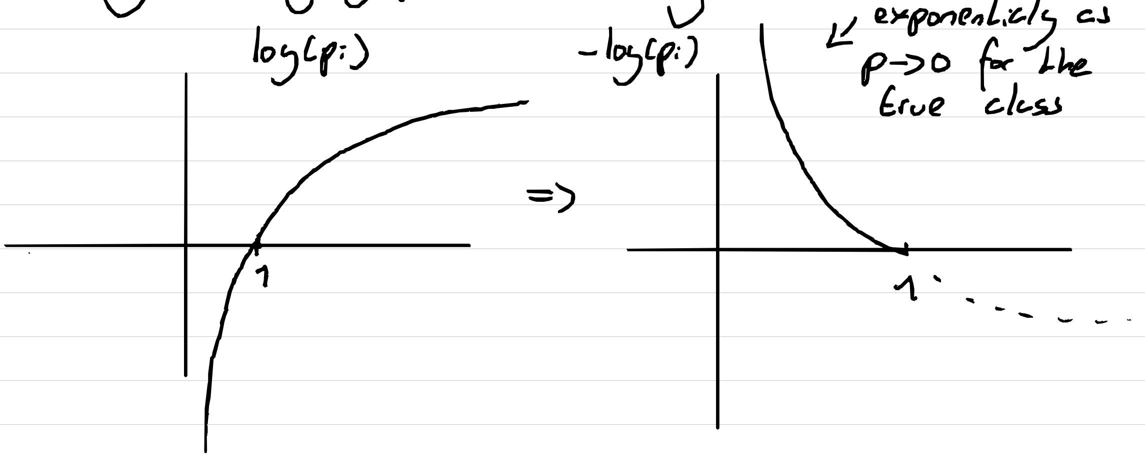
$$L = -\sum_{k=1}^{K} y_{ik} \log(p_{ik})$$

$y_i = \begin{bmatrix} 0 \\ 0 \\ 1 \\ \vdots \end{bmatrix}$ → Class $k = 3$

$y_{i3} = 1$, $y_{i2} = 0$

$$\text{Total loss: } L_T = -\sum_{i=1}^{N} \sum_{k=1}^{K} y_{ik} \log(p_{ik})$$

Why use $-y_i \log(p_i)$? Assume $y_i = 0$



We can use gradient descent to find θ.

$$L = -\sum_{i=1}^{N} \sum_{k=1}^{K} y_{ik} \log(p_{ik})$$ Recall $\theta_k = \begin{bmatrix} \theta_{1k} \\ \vdots \\ \theta_{jk} \\ \vdots \\ \theta_{d+1\,k} \end{bmatrix}$

$$\Rightarrow \frac{\partial L}{\partial \theta_{jk'}} = \frac{\partial}{\partial \theta_{jk'}} \left(-\sum_{i=1}^{N} \sum_{k=1}^{K} y_{ik} \log(p_{ik}) \right)$$

$$= -\sum_{i=1}^{N} \frac{\partial}{\partial \theta_{jk'}} \left(\sum_{k=1}^{K} y_{ik} \log(p_{ik}) \right)$$

$$= -\sum_{i=1}^{N} \sum_{k=1}^{K} y_{ik} \frac{\partial}{\partial \theta_{jk'}} \left(\log(p_{ik}) \right)$$

Log + chain rule

$\frac{d}{dx}(\log(x)) = \frac{1}{x}$

$\frac{d}{dx}(\log(u(x))) = \frac{1}{u} u'(x)$

$$\log(p_{ik}) = \log\left(\frac{e^{\underline{x}\theta_k}}{\sum_{c=1}^{K} e^{\underline{x}\theta_c}} \right)$$ $\log(e^x) = x$

$$= \log(e^{\underline{x}\theta_k}) - \log\left(\sum_{c=1}^{K} e^{\underline{x}\theta_c}\right)$$

$$= \underline{x}\theta_k - \log\left(\sum_{c=1}^{K} e^{\underline{x}\theta_c}\right)$$

$$\frac{\partial}{\partial \theta_{jk'}} \left(\log(p_{ik}) \right) = \frac{\partial}{\partial \theta_{jk'}} \left(\underline{x}\theta_k - \log\left(\sum_{c=1}^{K} e^{\underline{x}\theta_c}\right) \right)$$

$$= \frac{\partial}{\partial \theta_{jk'}} \left(\sum_{l=1}^{d+1} x_l \theta_{lk} - \log\left(\sum_{c=1}^{K} e^{\sum_{l=1}^{d+1} x_l \theta_{lc}}\right) \right)$$

$$= x_{ij} \delta_{kk'} - \frac{\partial}{\partial \theta_{jk'}} \left(\log \sum_{c=1}^{K} e^{\sum_{l=1}^{d+1} x_l \theta_{lc}} \right)$$

$\delta_{kk'} = 1$ if $k = k'$

$\delta_{kk'} = 0$ if $k \neq k'$

$$= x_{ij} \delta_{kk'} - \frac{1}{\sum_{c=1}^{K} e^{\sum_l x_l \theta_{lc}}} x_{ij} e^{\sum_{l=1}^{d+1} x_l \theta_{lc}}$$

$$= x_{ij} \delta_{kk'} - p_{ik'} x_{ij} = x_{ij}(\delta_{kk'} - p_{ik'})$$

$$\Rightarrow \frac{\partial L}{\partial \theta_{jk'}} = -\sum_{i} \sum_{k} y_{ik} x_{ij} (\delta_{kk'} - p_{ik'})$$

$$= \boxed{\sum_{i=1}^{N} x_{ij}(p_{ik'} - y_{ik'})}$$ → Can you prove why?

In vector form: $\nabla_\theta L = \underline{x}_i (p_i - y_i)^T$ → one sample

$\nabla_\theta L = X^T(P - Y)$ → multiple samples

Now, $\theta \leftarrow \theta - \alpha \nabla_\theta L$ where α is the step size

Exercises

Available on the notebook